DBMS.

PAGE No.
DATE 19 | 1 | 25

DBMS: structure where data is stored.

Data → collection of raw facts & figures. → qualitative (text) / quantitative (numeric)

↳ unorganised (no context given); can be stored anywhere in memory.

processed data is called INFORMATION. ie. meaning ful context.

Database: collection of interrelated data.

↳ whenever data is stored in structured form.

① central database
→ only single user can maintain and access database

② Distributed database.
→ multiple access & maintenance of database

• file organisation → method of arranging ... in computer memory.

when data is stored in manual or physical form → file base data
↓ backup.
duplication, no backup, less secure, inconsistency.
Eg: → excel, pdf, wordpad, notpad etc.

• DBMS: Soft / tool used to create / manage ... database.
↓ interface
or access.

① MySQL ② SQL ③ Oracle ④ MongoDB ⑤ Postgo SQL

① R DBMS ✓
relational DBMS
storing in form of table - row / column.

② NON ~~SQ~~ relation.
→ SQL executes.

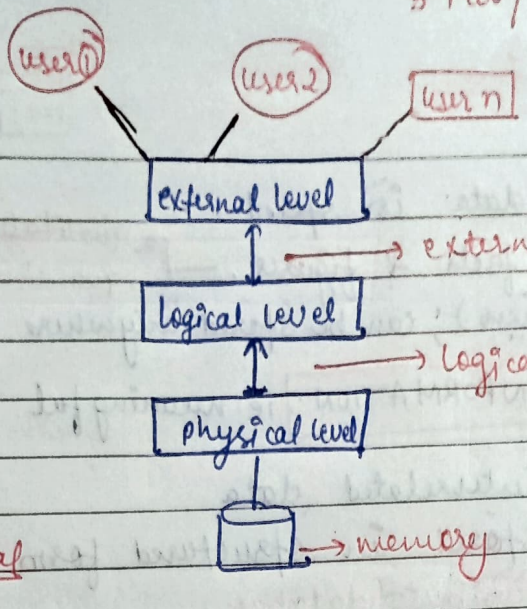
→ NO redundant, Security, backup, data is consistent, random access of data → features of DBMS. [Adv.]

Architect
(front end) → backend has DBMS;

• Application of DBMS: educat, hospitals, Bank, IRCTC, UPRTCSC, MakeMyTrip, etc.

Architecture of DBMS:

3 Tier / 3 view / ANSI Arch.



Objective of Architecture → multiple user accessing same database, diff customize views are maintained b/w users & hide actual actual implementation detail to user.

(SDV)

Objective of 3 Tier architecture —

1. It enables multiple users to access the same data with a personalized view while storing the same data only once.

2. Users of database should not worry about the physical implementation and internal working of databases such as data retrieval, data storing, etc.

(1) external level → maintenance of diff. customize views → at frontend.
or conceptual level.

(2) logical level - define structure for storage of data
Schema → blueprint / template / box for which data box storage is implemented
Logical representation of data base.
→ maintained at logical level.

(3) physical/internal level - storage of data in form of bits and byte & vice versa.

Mapping — to perform request of one level to another level.

2 type → ext to log. mapping & logical to physical mapping.

SIS. About 3 tier archi

→ describes structure for a specific database.

→ Separates the user applications & physical databases.

→ Mapping is used to transform the request b/w various database levels of architecture.

(1) External level: → highest level of data abstraction (to hide)
→ describes the part of entire database that a end-user concern.
→ diff. users need diff views of database and so that there can be many view level abstraction of same database.

(2) Conceptual / logical level: Next higher level of abstraction it describes what data are stored in database & what relationships are among those data.
↳ It is also called logical level / conceptual level because storage of data is defined logically at this level.

(3) Internal level: lowest level of data abstraction.
↳ describes how data are actually stored on storage devices. i.e your secondary storage or harddisc.

→ DBMS is maintaining all these levels.
↳ DBA → Database Administrators → maintaining & creating / logical level maintenance of database.

22/1/26

Database Schema —

Schema : template (skeletal) to represent & store data logically.

- Notation of schema = name of schema (column name – datatype [size])

example —

R.NO	Name	Sec	Age

Eg. Student sc (Rollno int, name char[20], Sec char(10), Age number(2));

→ schema doesn't contain data, only defines size of data to be stored.

→ DBA (Database administrator) defines schema.

→ when 1 schema – one schema, if more schema than relational schema.

(sir) → Database Schema –

- Structure that represents logical storage of data in a database.
- It represents organisation of data and provides info about the relationship b/w tables in a given database.

→ In actual, data is physically stored in files that may be in unstructured form but to retrieve it we need a structured form to do this, database schema is used.

- It provides knowledge about how data is organised in a database, & how it is associated with other data.

→ Schema does not contain data itself, instead it gives information regarding the shape & size of data.

→ complexity and size of schema vary as per size of the d.base

→ schema are of 2 types.

① logical schema

- works on logical level
→ where logical constraints that need to be apply for storing data.
→ constraints like integrity constraints, key constraints and referential constraints (set of rules & regulations)
→ logical schema is maintained at logical & conceptual level and designed by DBA.

② physical schema.

→ deals with how data is physically stored on storage devices like - disc, hard disc etc.
→ physical schema works or maintained at internal or physical level.

Database instance / snapshot of database —

→ Type of snapshot of an actual database as it exist at an instance of time.
- It is not fixed, it varies or can be changed frequently as per the operations performed.
→ Database instance can be changed by crude operation's. (create, read, update, delete)

Data Independency —

- change in one level is not affecting the other level.
→ i.e, freedom of levels.

Ex → Characteristic of being able to modify the schema

at one level of database without altering the schema at next level.

Logical data Independence - Change at conceptual or logical level without having change at the external level.
↳ Logical data Inde. occurs at user interface level.

Physical data Independence - Capacity to change the internal schema without having to change conceptual or logical schema.
↳ PDI occurs at logical ~~data~~ level interface level.

23/1/20

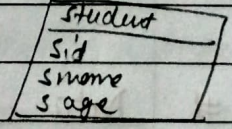
Data Models or Database Models

- conceptual
- representational → actual storage
  - ↳ Heirarchial DM
  - ↳ Network DM
- physical → Physical storage

(1) Heirarchial → maintain data in heirarchi → Object oriented DM
↳ tree like structure.
Disad → Cant maintain many relations, ~~struct~~

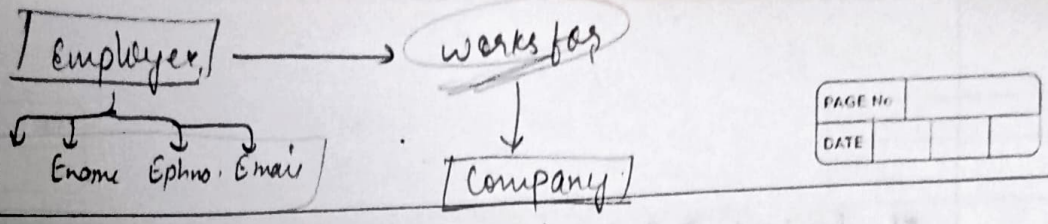
(2) Network → (graph) like structure → collection of node info in closed cct.
↳ child can have more than 1 parent unlike heirarchial.
Better than heirarchial.

(3) Object oriented → real world entity.



1. Conceptual model - Visualise the data.

Entity R/Ship Model.



PAGE No.
DATE

attributes → column in table

tupple is official name for row.

Sol →

- DB models are models that gives the idea that how final system will look like after its complete implementation.

→ Data Models is used to show how data is stored, connected, accessed and updated in DBMS.

→ Data Models provides us transparent picture of data which helps in creating an actual DB.

→ Data Models define data elements and relationship b/w Data and elements.

→ Data Models helps in representing data accurately.

→ Data models helps in finding missing data & also in minimizing data redundancy.

- Conceptual Model -

→ Used to conceptualize or visualize the data and its info for storage.

→ An entity relationship model also called ER Model.

ER Diagram :

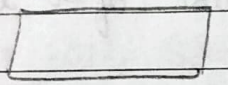
↳ used to define the relationship b/w entities present in a DB.

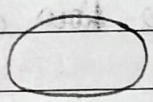
↳ Used for modelling of objects in real world.

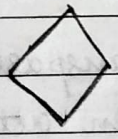
↳ standard sol' for visualizing data logically & also called GUI (Graphical user Interface) representation

of logical DB.

↳ Does not require deep knowledge of DBMS such as SQL to create ER Diag.

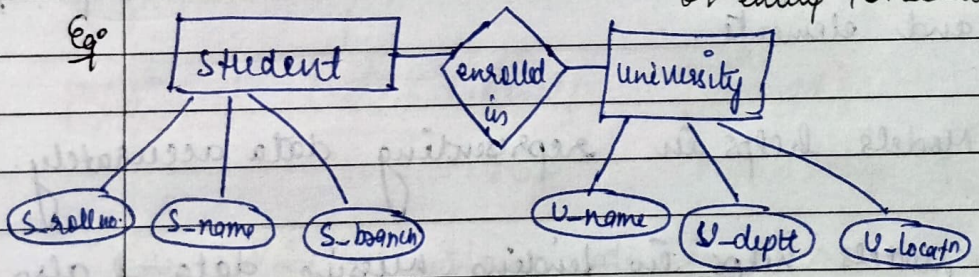

← entity


← attribute


← relationship

⟶ ← connecting entities or entity to relationship

Eg°



29/1/28

Types of attribute —

→ Simple / Single : holds single attribute & single value.

→ Multivalued : if attribute consists of more than 1 values.
↳ Notation = double circle : ◎

→ Derived - : which derives other attribute. Eg:- (DOB) → age.
Notation → doted circle : ⋯

→ Composite : Composite attribute. , which has many parts or types.

Eg → (Name) → first name
→ middle name
→ last name.

- Primary attribute → unique + not null

mapping constraints
- 1:M → single student → enrolled in single course.
- 1:M → single student enrolled in many course.
- M:1 → many students in one course.

M:M → many students in many courses

- cardinality — no. of time one entity participates in relationship
  1. 2 entities → Binary.
  2. single → unary.
  3. 3 entities → ternary.
  4. more than 3 → Nary.

① weak & strong entity set —
- strong → has prime attribute → single rectangle
- weak → has no prime attribute. → double rectangle.

Total & partial Participation —
- Total → every student registering for all course. → double line
- partial → every course regist. not mandatory.

Don't primary key used in other attribute then that is it is known as foreign key. and some values are used.

8. Design an ER diagram for a set of patients, to be treated by set of doctors, patients undergone medical test and tests report examined by set of doctors. establish relationship for above scenario, also define key attributes and types of relationship (mapping constraints).

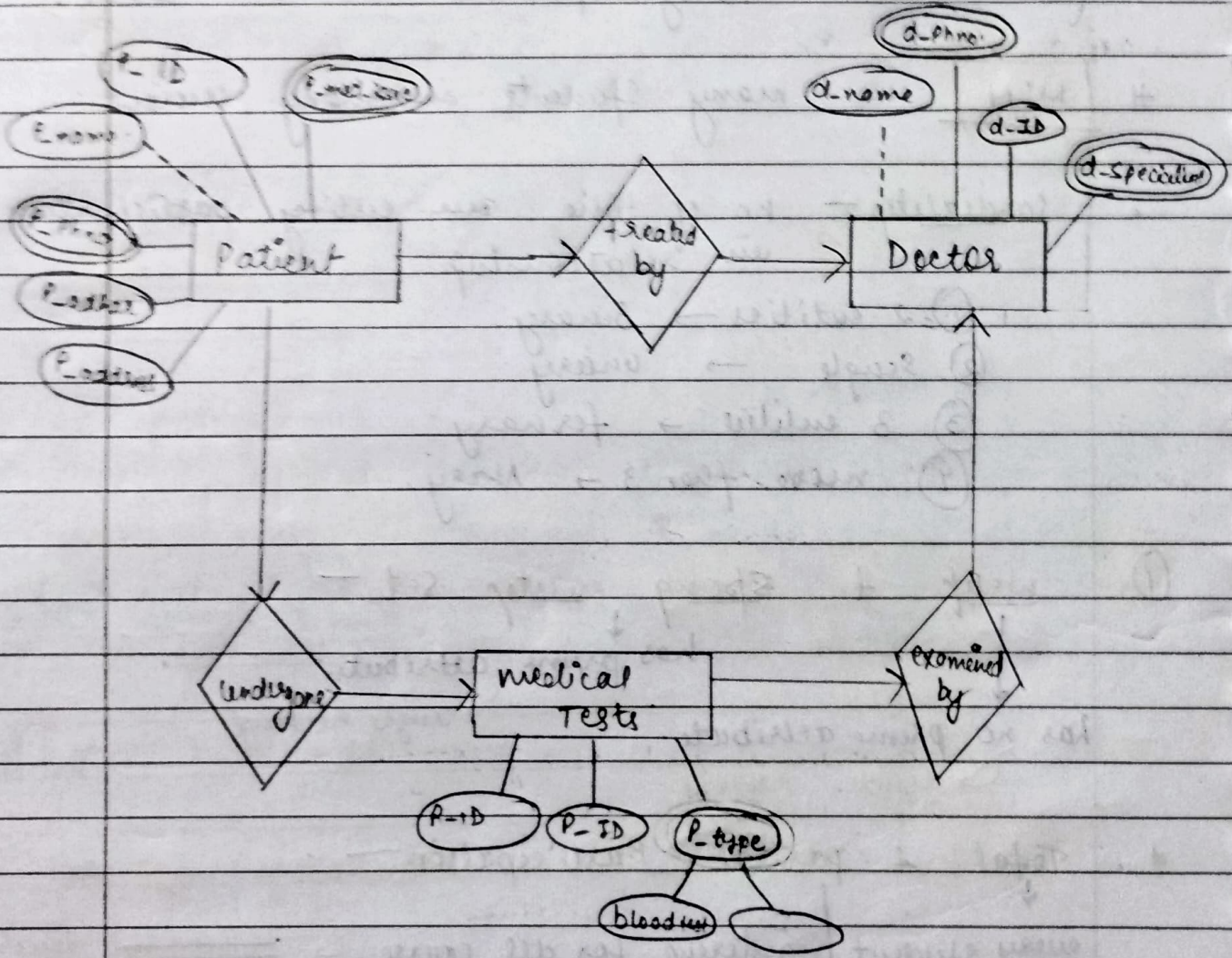
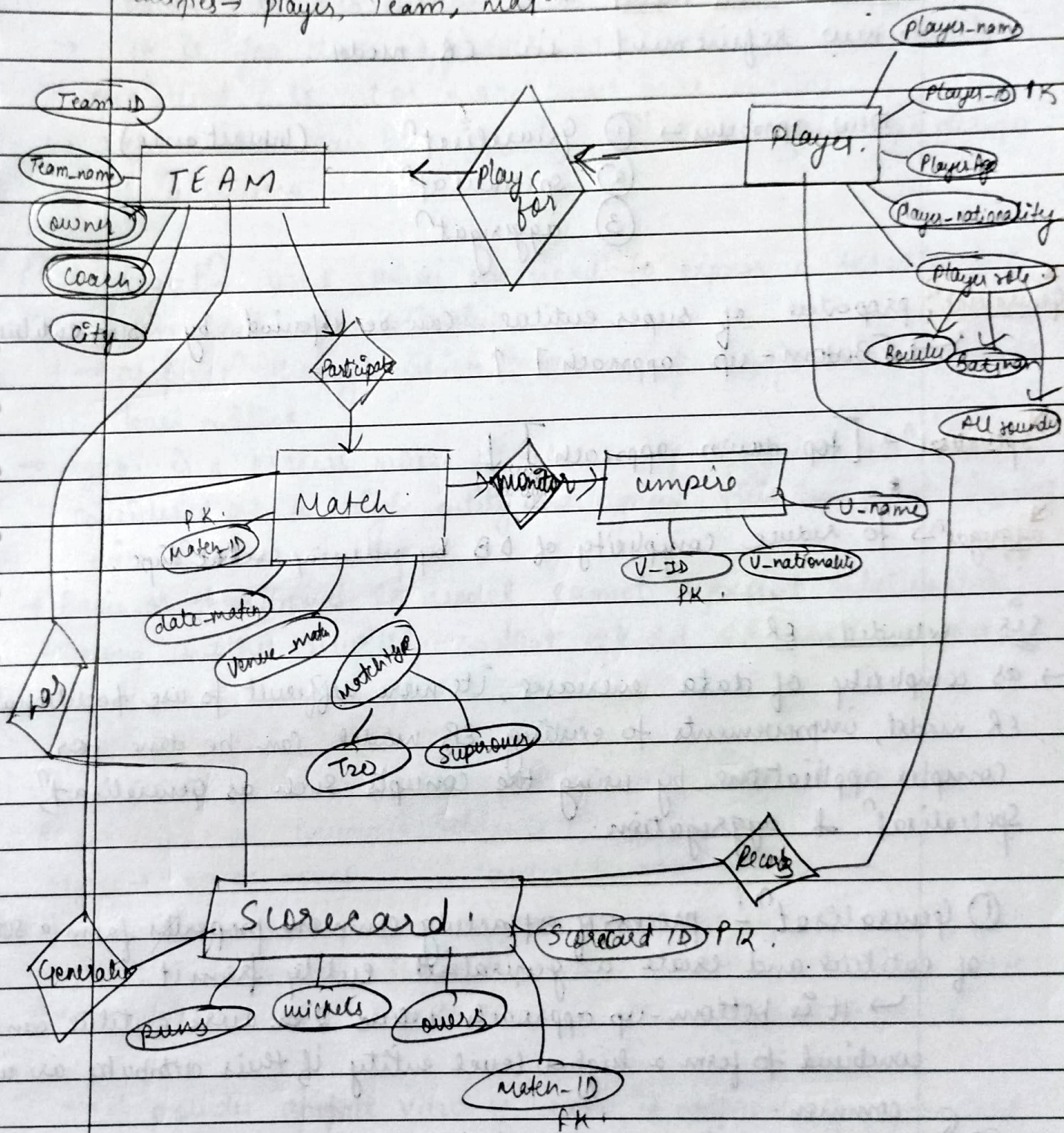


table →

Q. Consider a database which stores the details of cricket teams participating in IPL tournament to be scheduled in April 2026; establish the relationship for the above scenario and also identify mapping constraints & keys.

entities → players, Team, match.



30/1/26

PAGE NO
DATE

- extended ER model –

→ new refinement in ER model.

New approaches → ① Generalisat^n (Inheritance)
② specialisat^n.
③ aggregat^n

Generalisat^n: properties of super entities can be obtained by sub-entities
→ [Bottom-up approach.] ↑

Specialisat^n → [top down approach] ↓

aggregat^n → to reduce complexity of DB by reducing relat^nships.

Sir extended ER.

→ as complexity of data increases, its more difficult to use traditional ER model, improvements to existing ER model can be done for complex applications by using the concepts such as Generalisat^n, Specialisat^n & aggregation.

① Generalisat^n :- process of extracting common properties from a set of entities and create a generalised entity from it.
→ It is bottom-up approach where 2 or more entities can be combined to form a higher level entity if their attributes are in common.

② Concept of Inheritance - allow lower level entities i.e sub entities to inherit the attributes of higher level entity i.e, super entity.
→ Relationship b/w the lower level and higher level entity is depicted by an inverted triangle (▽) with a (-is-a) label to show the relat^ship.

③ Specialisat^n - opposite of generalisation, an entity is broken

down into sub entities based on their characteristic.
→ It is top down approach where higher level entity is specialised into 2 or more lower level entities.
↳ specialisation can be repeatedly applied to define the design of a scheme.

(3) Aggregat → used when we need to express a relationship among a set of relationship.
→ abstract through which relationship are treated as higher level entities.
→ aggr. is a process where a relationship b/w 2 entities is considered as a single entity and again this has a relationship with another entity.
→ Basic or traditional ER model cannot represent relationships among another relationship so that extended ER model is used.

- Relational Model —
→ used to demonstrate storage of data logically.

Row → tuple, record ; column → attribute
Degree → no. of rows ; cardiality → no. of columns.
Key → property used for defining attribute uniquely.
constraints - primary key, foreign key, default, check, not null, unique.
↳ traditional name → integrity, entity, key, Referential

adv.
→ It provides abstract view of data, it abstracts the physical structure from logical structure.
→ It represent how data is stored in relational databases. Relational
→ DB consists of collect of table, which consists of rows & columns to store data. table is called a relation & has assigned a unique name.
→ R.M. supports data independence.
→ RM helps user to use a query language i.e, SQL to query

Relation, attribute, Row / record / tuple, schema / Relational schema, degree, cardinality, Data item or value / keys, constraints.

PAGE NO
DATE

the database.